Predictive Machine Learning System for CyberArk Product Sales

DISSERTATION

Submitted in partial fulfillment of the requirements of the

Degree : MTech in Artificial Intelligence and Machine Learning

By

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2023AA05203

Under the supervision of

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DSECLZG628T / **AIMLCZG628T DISSERTATION**

Dissertation Title: Predictive Machine Learning System for CyberArk Product Sales

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Name of Student: Saya Venkata Thrivedi

ID No. of Student: 2023AA05203

Courses Relevant for the Project & Corresponding Semester:
1. Machine Learning
2. Artificial and Comp Intelligence
3. MLOPS
4. Deep Neural Networks

Abstract

In today's data-driven enterprise environment, the ability to predict customer purchase behavior is critical for designing targeted engagement strategies and improving conversion rates in high-stakes technology sales. This project proposes the development of a machine learning-based predictive analytics system to identify enterprise customers with a high likelihood of purchasing cybersecurity solutions, based on a diverse set of organizational, operational, and behavioral attributes captured during the sales lifecycle.

The dataset used in this study consists of structured records encompassing key business features such as company size, annual revenue, industry sector, security maturity, historical breach information, privileged user counts, identity tooling, cloud platform adoption, decision-maker roles, engagement scores, and lead source metadata. The overall objective is to build a robust classification system that accurately predicts the binary purchase outcome, enabling data-informed decision-making for sales and marketing teams.

The project will adopt a breadth-wise modeling strategy, beginning with the implementation of baseline classifiers such as logistic regression and decision trees to establish foundational performance metrics. These will be followed by more advanced ensemble methods, including Random Forest, Gradient Boosting (e.g., XGBoost), and CatBoost, which are well-suited for capturing complex feature interactions. Finally, the study will explore the application of neural network architectures to assess their efficacy in modeling non-linear dependencies in high-dimensional enterprise sales data.

Comprehensive evaluation will be conducted using metrics such as precision, recall, F1-score, and ROC-AUC, along with model interpretability techniques such as SHAP to provide actionable insights into the key drivers of purchase decisions. The best-performing model

will be selected based on a combination of predictive performance, generalizability, and interpretability.

The solution will be deployed through a lightweight, interactive user interface to facilitate real-time inference. The complete pipeline, including model training, evaluation, and deployment, will be containerized using Docker and integrated with CI/CD tools to support continuous development and delivery.

Over the span of sixteen weeks, this project will deliver a scalable and interpretable machine learning solution, combining rigorous model experimentation with best practices in MLOps, thus demonstrating the applicability of predictive modeling to enterprise sales forecasting within the cybersecurity domain.

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI
II SEMESTER 24-25
DSECLZG628T / **AIMLCZG628T DISSERTATION**
Dissertation Outline (Abstract)

BITS ID No. 2023AA05203 **Name of Student:** Saya, Venkata Thrivedi

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Qualification and Experience: Bachelor of Technology in Electronics and Communication Engineering

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Topic of Dissertation: Predictive Machine Learning System for CyberArk Product Sales

(Signature of Student) (Signature of Supervisor)

Date:---------- Date:----------

Broad Area of Work

The project falls under the broad domain of Artificial Intelligence (AI) and more specifically focuses on the subfields of Applied Machine Learning (ML), Predictive Modeling, and Enterprise Decision Intelligence. In today's data-driven ecosystem, organizations increasingly rely on machine learning systems to make informed strategic decisions, including the prediction of customer intent and purchase behavior. This project contributes to this broader vision by applying supervised learning techniques to predict the likelihood of cybersecurity product purchases by enterprises based on their digital and organizational characteristics.

The work intersects multiple thematic areas including:

- *Business Intelligence and Predictive Analytics:* Leveraging historical and real-time enterprise-level data to forecast potential purchasing intent.
- *Model Evaluation and Selection*: Implementing and benchmarking classical, ensemble-based, and neural network models for binary classification tasks.
- *MLOps and Model Lifecycle Management*: Integrating DevOps practices into the ML pipeline to ensure reproducibility, scalability, and maintainability of the deployed solution.
- *Human-Centered AI Interfaces:* Delivering model predictions through a simple, user-friendly web interface for decision support.

This interdisciplinary project blends concepts from computer science, statistics, and enterprise sales operations to build a robust and explainable ML-driven recommendation system that can assist cybersecurity product teams in identifying high-value customer prospects efficiently.

Objectives

The key objectives of this project are as follows:

- To explore and preprocess enterprise sales data capturing organizational attributes, security readiness, and engagement metrics.
- To design and implement a suite of predictive models (baseline, advanced, and neural network-based) for forecasting the likelihood of cybersecurity product purchases.
- To systematically compare the performance of models using appropriate classification metrics (Precision, Recall, ROC-AUC, F1 Score).
- To select and fine-tune the best-performing model for deployment.
- To build a simple, interactive user interface for prediction demonstration.

- To containerize the solution using Docker and implement CI/CD for automated retraining and deployment.

Scope of Work

The scope of this dissertation includes the end-to-end development of a machine learning pipeline to solve a real-world problem in enterprise product recommendation and intent prediction. The project involves:

- Preprocessing a diverse, real-world styled dataset containing enterprise, security, and engagement attributes.
- Building and evaluating predictive models such as Logistic Regression, Random Forest, Gradient Boosting, and Feedforward Neural Networks.
- Designing a UI for interacting with the final predictive model.
- Enabling model reproducibility and deployment using containerization (Docker) and optionally integrating CI/CD pipelines for model retraining and monitoring.

Detailed Plan of Work (Sample) (for 16 weeks)

Serial Number of Task/Phases	**Tasks or subtasks to be done** (be precise and specific)	Start Date-End Date	Planned duration in weeks	Specific Deliverable in terms of the project
1	Requirement analysis and data understanding	Week 1	1	Requirement doc, data schema report
2	Data cleaning, exploratory data analysis (EDA), data validation	Week 2-3	2	Dataset Cleaning and EDA Report
3	Feature engineering and data preprocessing (scaling, encoding, etc.)	Week 4-5	2	Data Preprocessing
4	Implementation of baseline models (Logistic Regression, Decision Tree)	Week -6	1	Baseline model results
5	Implementation of advanced models (Random Forest, XGBoost, etc.)	Week 7-8	2	Advanced model results
6	Implementation of neural network models (MLP, tuning, etc.)	Week 9	1	Neural model results
7	Model comparison and selection using metrics and cross-validation	Week 10- 11	1	Model Evaluation report
8	Model Export & UI development (simple web interface for predictions)	Week 12	1	Working UI integrated with best model
9	Dockerization of the application	Week 13	1	Docker container for complete app
10	CI/CD pipeline design and automation (GitHub Actions / Jenkins)	Week 14	1	CI/CD pipeline design and automation (GitHub Actions / Jenkins)
11	Final testing, documentation, and presentation preparation	Week 15 - 16	2	Final report and presentation

Literature References

The following are referred journals from the preliminary literature review.

[1] Aurélien Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, O'Reilly Media, 2019.

[2] Choudhury, T., *Practical MLOps*, O'Reilly Media, 2021.

[3] Ribeiro, M. T., Singh, S., & Guestrin, C., *"Why Should I Trust You?": Explaining the Predictions of Any Classifier*, KDD, 2016.

Supervisor's Rating of the Technical Quality of this Dissertation Outline

EXCELLENT / GOOD / FAIR/ POOR (Please specify): _____

Supervisor's suggestions and remarks about the outline (if applicable).

Date 21-May-2025

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(Signature of Supervisor)

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